

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title:

Multi-Language Translator Web Application Using NLP

1. Introduction

This document specifies the requirements for a Multi-Language Translator Web Application developed using Python Flask and NLP.

2. Purpose

To provide accurate translation from English to Hindi and support future languages.

3. Scope

Users enter English text and receive translated output using Helsinki-NLP transformer model deployed on Render.

4. Technologies Used

Frontend: HTML, CSS, JavaScript

Backend: Python Flask

Model: Helsinki-NLP/opus-mt-en-hi

Deployment: Render

5. Functional Requirements

FR1: Accept English text

FR2: Select target language

FR3: Translate using NLP model

FR4: Display output

FR5: REST API /translate

FR6: Error handling

FR7: Cloud deployment

6. Non Functional Requirements

Performance: ≤ 3 seconds response

Usability: Simple UI

Reliability: High uptime

Security: Input validation

Scalability: Multi-user support

7. Architecture Flow

User -> Frontend -> Flask API -> NLP Model -> Response

8. API Example

POST /translate

Input: Good Morning

Output: ■■■■■■■■

9. Deployment

GitHub connected to Render

Gunicorn server

requirements.txt

10. Future Enhancements

Multiple languages

Speech input

Mobile UI

Translation history

Authentication

Conclusion

This project demonstrates NLP based machine translation using Flask and Render.